

1.What is data cleaning, and why is it important in data analysis?

What are the potential consequences of analysing unclean or messy data?

Explain the common steps involved in cleaning and organizing data.

Answer: Data cleaning is the process of identifying and correcting errors or inconsistencies in data to improve its quality and reliability for analysis. This process involves removing inaccuracies, ensuring that data is uniform, and filling in missing values. Data cleaning is a crucial part of the data preparation phase, as it ensures that the datasets used for analysis are accurate, complete, and suitable for the intended purpose.

Clean data provides accurate insights and outcomes, which is essential for making informed business or research decisions. Errors in data can lead to misleading results, impacting conclusions and strategies.

Common steps involved in data cleaning are:-

Data Profiling: Analyze the dataset to understand its structure, content, and relationships. This includes checking for types of variables, missing values, and outliers.

Removing Duplicate Data: Identify and eliminate any duplicate entries to ensure that each record is unique and prevents inflated analysis.

Handling Missing Values: Determine how to address missing data, whether through imputation methods (filling in missing values using statistical techniques), deletion, or flagging for further review.

Standardizing Data: Convert data into a uniform format. This includes standardizing date formats, text casing, and categorical variable names to ensure consistency throughout the dataset.

Filtering Outliers: Identify and assess outliers that may distort analysis. Depending on context, one might choose to remove, correct, or keep them for further examination.

Validating Data: Check the accuracy of data against external sources or predefined rules to ensure it is correct.

Transforming Data: Create new variables or change the format to make data more suitable for analysis, such as normalizing numerical values or encoding categorical variables.

Documentation: Maintain records of all cleaning processes and decisions. This documentation aids transparency and replicability for other analysts.

Final Review: Conduct a final assessment of the cleaned data before proceeding with the analysis to confirm it meets the necessary quality standards.

3. Explain the use of text functions such as TRIM, LEFT, RIGHT, MID, and CONCAT in data cleaning.

Answer: -

TRIM Function Removes any leading and trailing space from a text string.

LEFT is useful when you need to isolate a prefix or a specific portion of text. For instance, if you have a column of codes, and you want to extract the first few characters of a code for categorization, you can use LEFT.

The RIGHT function extracts a specified number of characters from the end of a text string.

The MID function extracts a substring from a text string, starting at a specified position and continuing for a specified number of characters.

The CONCAT function combines multiple text strings into one single string.

4. What is the role of date functions like TODAY in managing datasets?

Answer: -

Date function such as today are crucial in managing data sets especially in data analysis, reporting, time-based calculations as it provides the current date automatically without the need for manual update. This is particularly useful when creating reports or dashboard that need to reflect real time data.